

part of TUM THINK TANK

QuantWorld Hackathon

Quantum optimization with QUBO

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The Wiesn Wobble Problem

You have time for two stops. Which pair would you choose?



Beer tent



Carousel



Autoscooter

Beer tent https://commons.wikimedia.org/wiki/File:InsideAugustinerTentOktoberfest.jpg
Carousel https://commons.wikimedia.org/wiki/File:Oktoberfest_Karussell.JPG
Autoscooter https://commons.wikimedia.org/wiki/File:Taunus_Wunderland_Autoscooter.JPG

OPENING EXAMPLE

Three stops, two choices

This fictional visitor assigns enjoyment points to each stop.

B

BEER TENT

8 points

C

CAROUSEL

6 points

A

AUTOSCOOTER

5 points

The visitor can choose exactly two stops.



Beer before a spin?

Our visitor plans the beer tent before the carousel. Spinning then costs 4 points if both are chosen.

Penalty applies only when both beer tent and carousel are chosen.

Beer tent + Carousel	$8 + 6 - 4 = 10$
Beer tent + Autoscooter	$8 + 5 - 0 = 13$
Carousel + Autoscooter	$6 + 5 - 0 = 11$

LECTURE MAP

Today: from problem to evidence

1

01: Context

Quantum landscape and background

2

02: Model

QUBO variables and constraints

3

03: Solve

Classical and quantum methods

4

04: Evaluate

Benchmark and practical workflow

01 Foundations

Quantum methods in context before we build a model.

Algorithms at a glance

Grover's algorithm

Quantum phase estimation

Amplitude amplification

Simulated annealing

VQE

Quantum annealing

QAOA

Simon's algorithm

Tabu search

Deutsch–Jozsa

Shor's algorithm

Bernstein–Vazirani

Quantum amplitude estimation

Our focus this lecture: optimization

Grover's algorithm

Quantum phase estimation

Amplitude amplification

Simulated annealing

VQE

Quantum annealing

QAOA

Simon's algorithm

Tabu search

Deutsch–Jozsa

Shor's algorithm

Bernstein–Vazirani

Quantum amplitude estimation

A caveat about quantum advantage

CAVEAT

> There is no established quantum advantage for quantum optimization.

Results depend on the problem class, the classical baseline and which costs enter the clock.

USEFUL NOW

The same modelling skills support exact solvers, mixed integer optimization and classical heuristics.

Formulation, verification and benchmarking remain valuable even when the quantum solver does not win.

Your project may fit other quantum algorithms with proven theoretical speedups.

Grover offers for example a quadratic query speedup. Practical runtime gains are not guaranteed.

How proficient are you with gate model quantum computing?



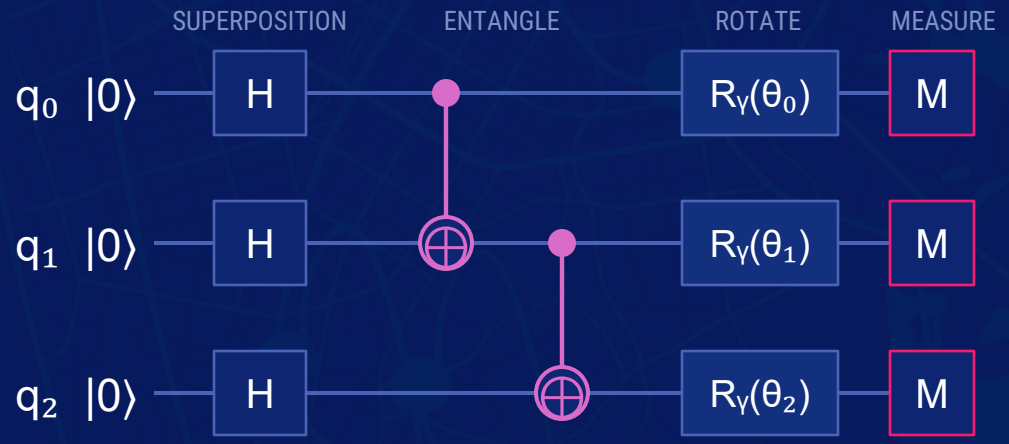
0	1	2	3	4
Never used one	Can read a circuit	Can build simple circuits	Have used a quantum SDK	Can implement an algorithm

Raise your hand for the highest level that fits

Gate model and quantum annealing

GATE MODEL

Specify operations



Ordered operations on named qubits

QUANTUM ANNEALING

Specify an energy landscape

A Hamiltonian assigns an energy to every configuration of the qubits. The lowest one is the ground state.

$$H(s) = A(s)\, H_0 + B(s)\, H_P$$

H_0 starting Hamiltonian, its ground state is easy to prepare

H_P problem Hamiltonian, its ground state is the solution

$A(s)$, $B(s)$ the annealing schedule, A falls while B rises

Sampling returns low energy states

Optimization as a ground state problem

CANDIDATE STATES

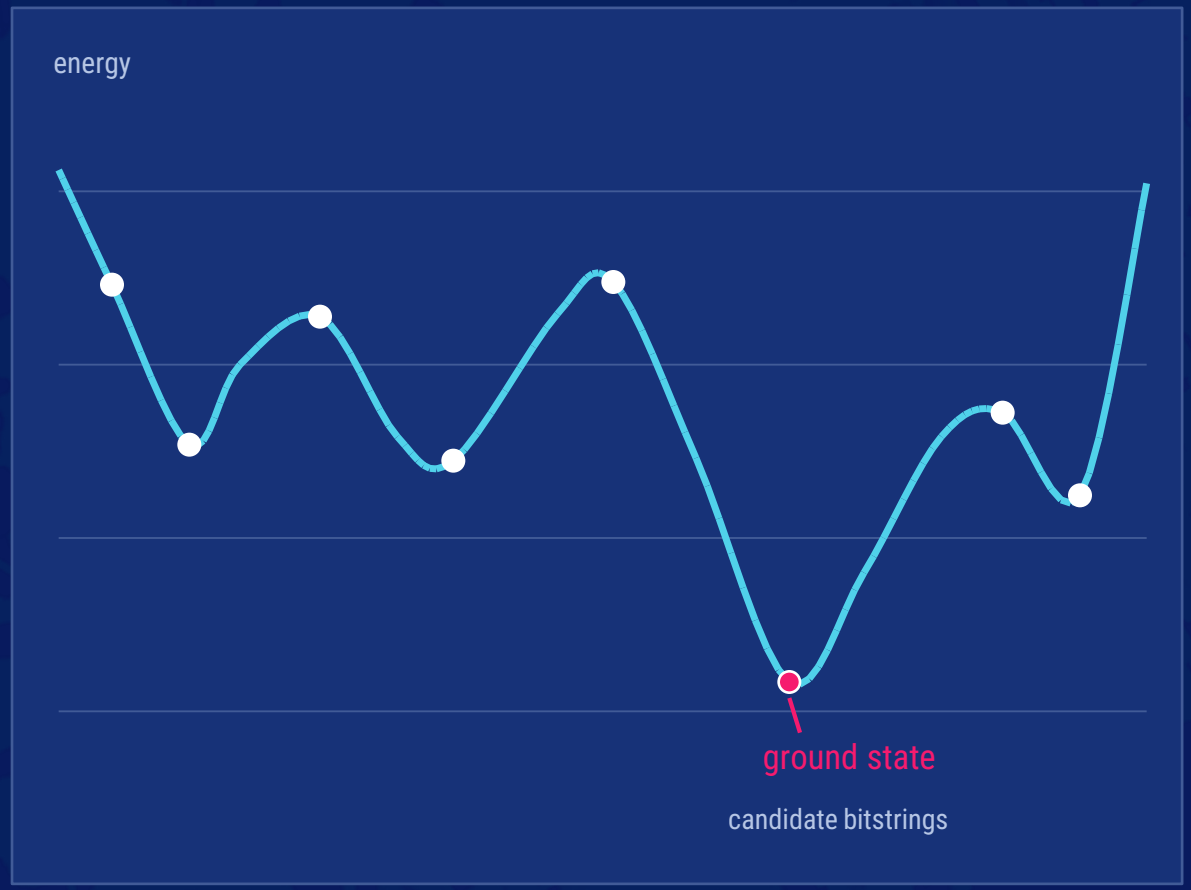
Every bitstring x represents one candidate solution.

ENERGY

The problem Hamiltonian assigns an objective value to each candidate.

$$H_P(x) = E(x)$$

The ground state of H_P encodes the optimum.



Have you ever formulated an optimization problem?



0	1	2	3	4
Never	Have had a lecture on optimization	Written an objective and constraints	Built a QUBO or Ising model	Validated models in practice

Raise your hand for the highest level that fits

02

QUBO modelling

Translate decisions, objectives and rules into energy.

QUBO

Q **QUADRATIC** items interact in pairs at most

U **UNCONSTRAINED** hard constraints become energy penalties

B **BINARY** x_i is 0 or 1

O **OPTIMIZATION** find the lowest value

$$\min \quad E(x) = x^T Q x$$

$$x_i \in \{0, 1\}$$

One sentence per variable

$x_i = 1$ means asset i is selected

Lecture convention: lower energy is better. Some tools and papers maximize, so check the sign.

The Wiesn Wobble QUBO

Select exactly two stops. Choosing beer tent and carousel together costs 4 points.

BINARY CHOICES

Beer tent	x_B	8
Carousel	x_C	6
Autoscooter	x_A	5

x = 1 if that stop is chosen, otherwise 0

$$\text{maximize } 8x_B + 6x_C + 5x_A - 4x_Bx_C$$

$$\text{subject to } x_B + x_C + x_A = 2$$

$$\min E(x) = -8x_B - 6x_C - 5x_A + 4x_Bx_C + 10(x_B + x_C + x_A - 2)^2$$

UPPER TRIANGULAR Q

$$x = (x_B, x_C, x_A)^T$$

$$Q = \begin{bmatrix} -38 & 24 & 20 \\ 0 & -36 & 20 \\ 0 & 0 & -35 \end{bmatrix}$$

$$E(x) = x^T Q x + 40$$

Linear terms on the QUBO diagonal

$$x_i^2 = x_i \quad \text{for } x_i \in \{0,1\}$$

$$a_i x_i = a_i x_i^2$$

Therefore a_i goes into Q_{ii}

A pair term $b_{ij}x_ix_j$ contributes only when both decisions i and j are active.

	x_1	x_2	x_3
x_1	a_1	b_{12}	b_{13}
x_2		a_2	b_{23}
x_3			a_3

diagonal = individual terms

off diagonal = pair terms

We store each pair once in the upper triangle because $x_ix_j = x_jx_i$.

Portfolio Objective: choose two of four assets

Asset	Expected return μ_i
A	1.20
B	0.90
C	0.70
D	0.50

Covariance matrix Σ

	A	B	C	D
A	0.40	0.18	0.12	0.10
B	0.18	0.30	0.08	0.07
C	0.12	0.08	0.25	0.06
D	0.10	0.07	0.06	0.20

Diagonal: variance of one asset Off diagonal: covariance between every asset pair

The portfolio objective

reward risk

$$\min \quad f(x) = -\mu^T x + \lambda x^T \Sigma x$$

select high return assets

avoid risky combinations

$x_i = 1$ means asset i is selected

For the example instance

$\lambda = 0.8$

Constraint

choose exactly K = 2 assets

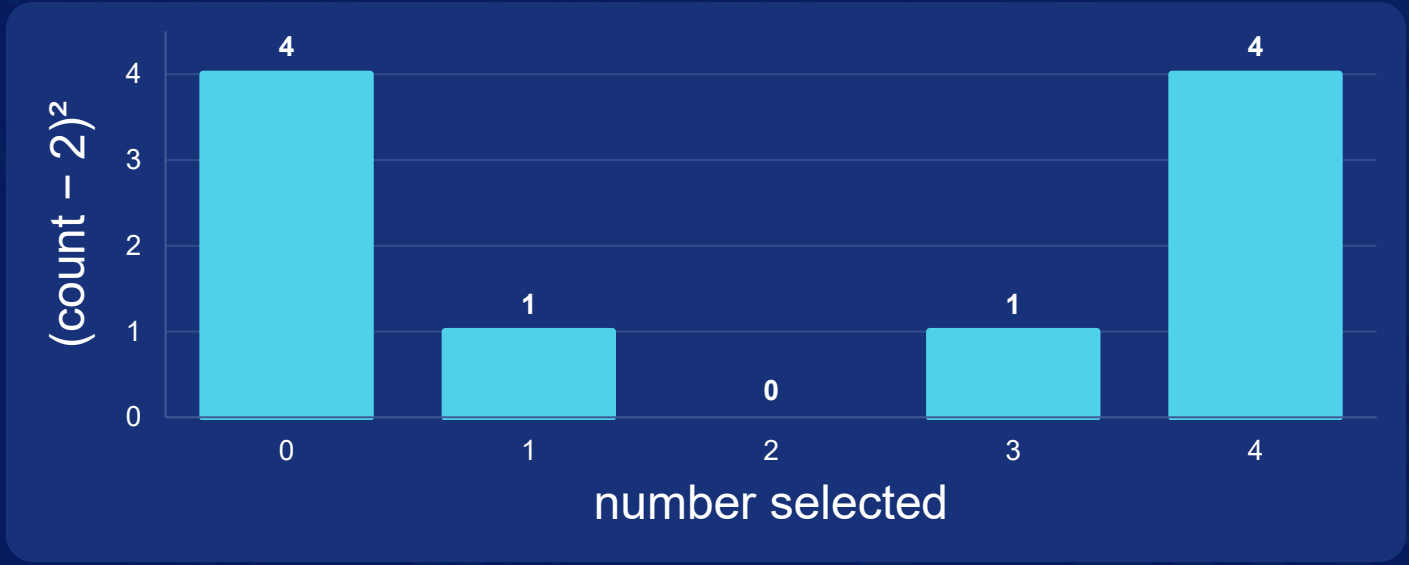
Cardinality as an energy penalty

$$\sum_i x_i = K$$

hard rule

$$P\left(\sum_i x_i - K\right)^2$$

energy penalty



Only count = 2 has zero penalty

The solver may still accept a violation when the objective gain is larger than the added penalty.

Expanding the penalty

$$P\,(\sum_i x_i - K)^2$$

$$= P\,(\sum_i x_i)^2 - 2PK \sum_i x_i + PK^2$$

$$= P(1 - 2K) \sum_i x_i + 2P \sum_{i<j} x_i x_j + PK^2$$

diagonal

off diagonal

constant offset

Assembling the QUBO matrix

$$Q_{ii} = -\mu_i + \lambda\Sigma_{ii} + P(1 - 2K)$$

$$Q_{ij} = 2\lambda\Sigma_{ij} + 2P \quad \text{for } i < j$$

$\lambda = 0.8$ $K = 2$ $P = 0.3$

offset = $PK^2 = 1.2$

Keep the offset with the model. Dropping it changes reported energy values even when the best bitstring stays the same.

Q	A	B	C	D
A	-1.780	0.888	0.792	0.760
B		-1.560	0.728	0.712
C			-1.400	0.696
D				-1.240

Reading the QUBO matrix

−1.780	+0.888	+0.792	+0.760
	−1.560	+0.728	+0.712
		−1.400	+0.696
			−1.240

DIAGONAL

Individual reward, cost or bias

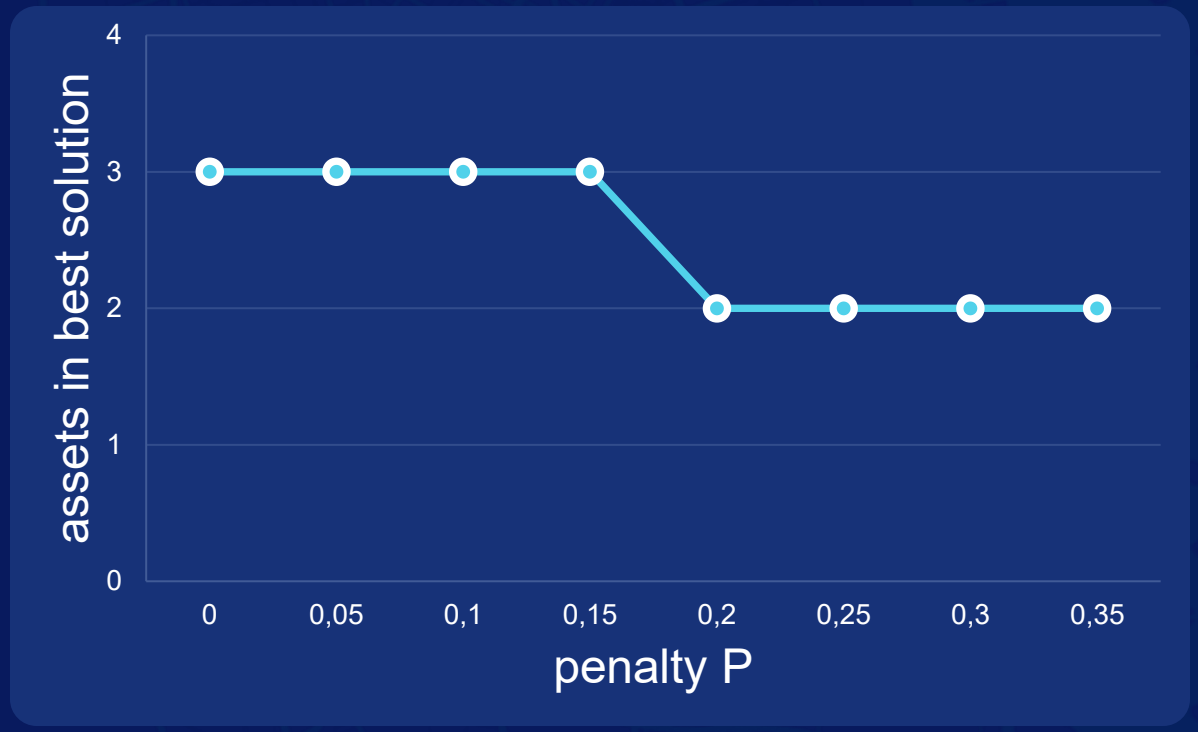
The teaching matrix has negative diagonals. Selecting an asset lowers energy before pair effects and penalties are considered.

OFF DIAGONAL

Pairwise interaction

Positive pair values raise energy when both decisions are one. Covariance and the cardinality penalty both contribute here.

Penalty strength: feasibility versus scale





First feasible point shown: P = 0.20

Large P compresses the objective signal

Exact enumeration of the four asset example

The penalty recipe: square the violation

HARD CONSTRAINT: $\sum_i x_i = K$ a solution x' that satisfies it is called feasible

1 WRITE THE RULE AS A VIOLATION

$$g(x) = \sum x_i - K$$

g measures how far an assignment sits from the rule. A feasible x' gives g(x') = 0. A larger value means a worse break.

2 SQUARE IT

$$g(x)^2 \geq 0, \quad \text{and} \quad g(x')^2 = 0 \quad \Leftrightarrow \quad x' \text{ is feasible}$$

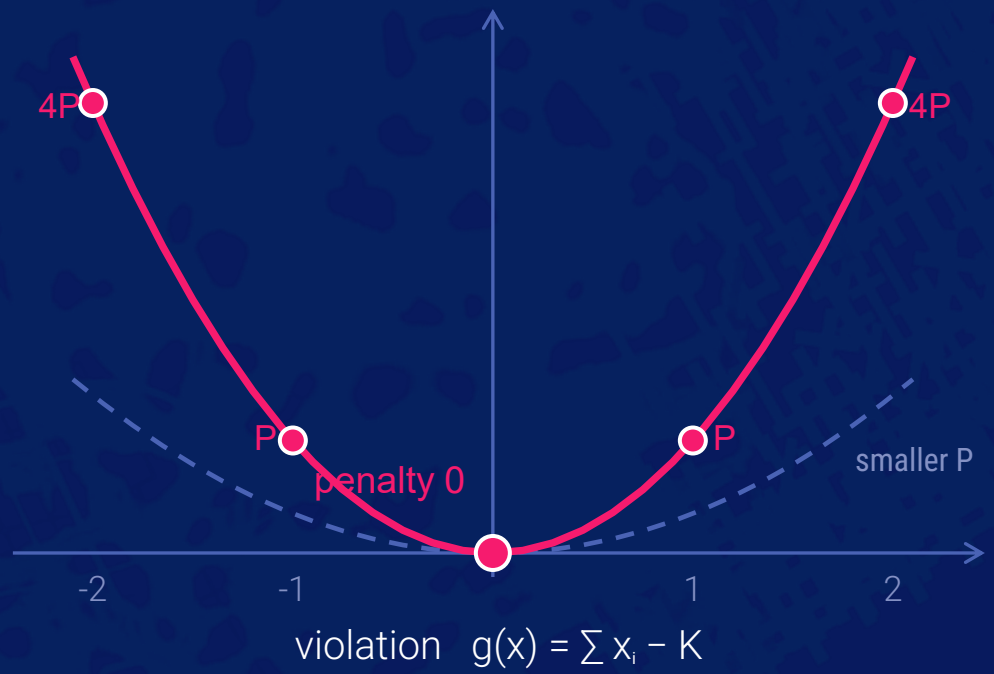
The square is zero on exactly the feasible assignments x' and positive on every other one. Without it, breaking the rule would lower the energy.

3 SCALE IT AND ADD IT TO THE OBJECTIVE

$$\min \quad f(x) + P\, g(x)^2$$

P only sets how steep the walls are. It never moves the zero, so a feasible solution keeps the energy the objective gave it.

PENALTY AS A FUNCTION OF THE VIOLATION



Common constraint patterns

Rule	Classical form	Penalty
Exactly K selected	$\sum x_i = K$	$P\,(\sum x_i - K)^2$
Exactly one of a group	$\sum_{i \in G} x_i = 1$	$P\,(\sum_{i \in G} x_i - 1)^2$
Not both i and j	$\neg(x_i \wedge x_j)$	$P\,x_i x_j$
If i then j	$x_i \Rightarrow x_j$	$P\,x_i(1 - x_j)$
General equality	$\sum w_i x_i = C$	$P\,(\sum w_i x_i - C)^2$
At most B	$\sum w_i x_i \leq B$	$P\,(\sum w_i x_i + s - B)^2$

Pure BQM

Constraints become penalties inside one unconstrained objective.

Hybrid CQM

Equality and inequality constraints can remain explicit.

Slack s needs its own binary encoding on a BQM or QPU.

Translation costs

CONTINUOUS QUANTITY

b bits precision ≈ 2^{-b}

$w \in [0, 10] \quad \rightarrow \quad w = (10/15)\,(y_0 + 2y_1 + 4y_2 + 8y_3)$

Four bits give steps of 0.67. Each extra bit halves the step.

INEQUALITY

$\log_2$ range bits slack variables

$\sum_i x_i \le 3 \quad \rightarrow \quad \sum_i x_i + s = 3 \quad \text{with} \quad s = s_0 + 2s_1$

Two bits cover a slack from 0 to 3. The equality is squared as before.

CUBIC TERM

auxiliary variable plus a penalty

$x_i\, x_j\, x_k \quad \rightarrow \quad x_i\, y \qquad \text{with} \quad y = x_j\, x_k$

enforced by $P\,(x_j\, x_k - 2\, x_j\, y - 2\, x_k\, y + 3\, y)$

Zero exactly when $y = x_j\, x_k$, positive otherwise. One extra bit per cubic term.

A useful warning sign

Most variables no longer represent real decisions

If auxiliary bits and penalty terms dominate the model, reconsider the formulation or keep the original classical model.

Equivalent QUBO and Ising formulations

QUBO

$x_i \in \{0, 1\}$

asset out / asset in

The modelling side. Variables read as yes or no decisions.

$s_i = 2x_i - 1$

$x_i = (s_i + 1) / 2$

ISING

$s_i \in \{-1, +1\}$

spin down / spin up

The machine side. The form a real annealer is built around.

$$H_P = \sum_i h_i s_i + \sum_{i<j} J_{ij} s_i s_j + \text{offset}$$

Ocean: bqm.to_ising()

Binary and Ising variables encode different relations

BINARY: x in {0, 1}

x_i	x_j	$x_i \times x_j$	meaning
0	0	0	neither
0	1	0	only j
1	0	0	only i
1	1	1	both true

The product is 1 only when both decisions are true.

ISING: s in {-1, +1}

s_i	s_j	$s_i \times s_j$	meaning
-1	-1	+1	same
-1	+1	-1	opposite
+1	-1	-1	opposite
+1	+1	+1	same

The product is positive for equal spins and negative for opposite spins.



03 Solvers

Several ways to search the same model.

One model, several solver approaches

We compare classical, hybrid and quantum approaches. QUBO is a model, not an algorithm.

CLASSICAL

Simulated annealing

Temperature allows occasional uphill moves.

CLASSICAL

Tabu search

Memory prevents immediate revisits.

QUANTUM

Quantum annealing

A changing Hamiltonian searches low energy states.

HYBRID

QAOA

A classical optimizer tunes a quantum circuit.

ONLY A SMALL SELECTION

Genetic algorithms, particle swarm, constraint programming and mixed integer solvers also solve these problems as well.
The right choice follows from the problem, not from the model.



03.1

Simulated annealing

A classical search that sometimes accepts worse moves.

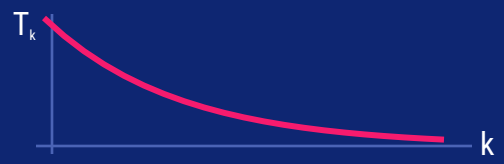
Simulated annealing: the classical counterpart

SIMULATED ANNEALING

```
input   QUBO energy E(x), schedule T_k
x       ← initial bitstring
x_best  ← x

for k = 1, …, K
    x'  ← flip one bit of x
    ΔE  ← E(x') − E(x)
    accept x' if ΔE ≤ 0
    otherwise accept with probability
        exp(−ΔE/T_k)
    update x_best

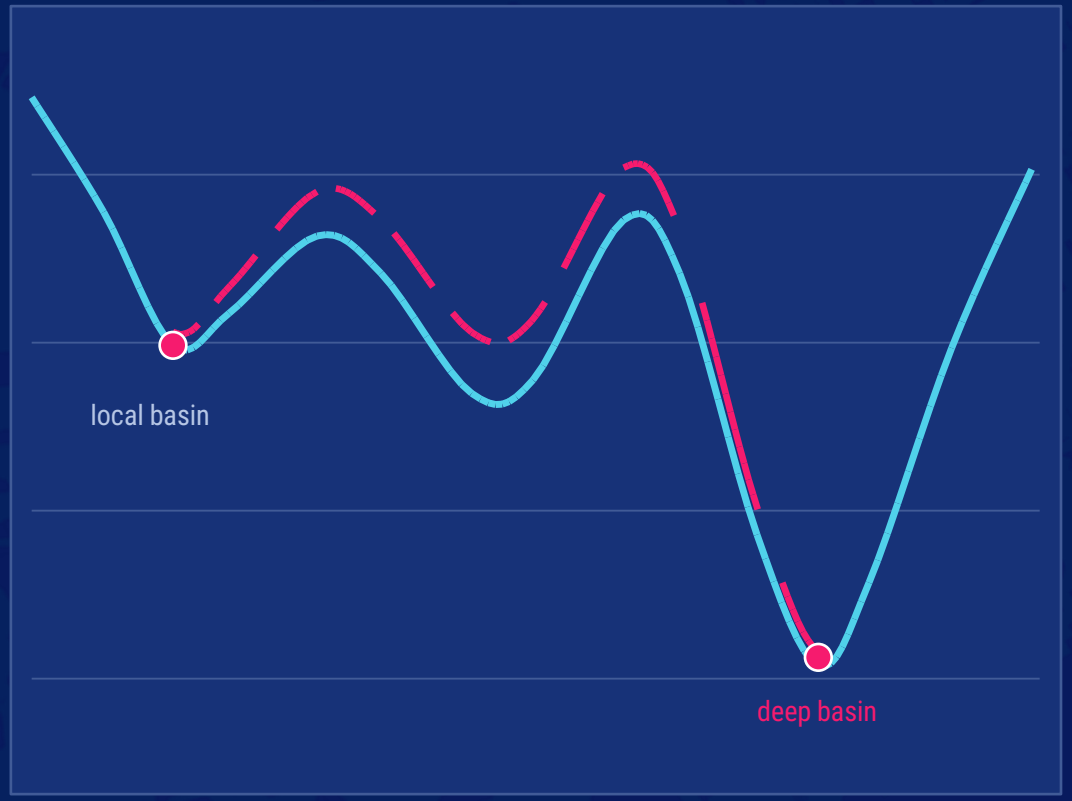
return x_best
```



COOLING SCHEDULE

Temperature falls as k grows, so uphill moves get rare.

Energy landscape



High temperature permits uphill moves. Cooling makes them less likely.



03.2

Tabu search

A classical search with short term memory.

Tabu search: memory instead of temperature

TABU SEARCH

```
input   QUBO energy E(x), tenure L
x       ← initial bitstring
x_best  ← x
tabu_until[i] ← 0 for every bit i

for k = 1, …, K
    allowed ← { i : tabu_until[i] < k }
    x'      ← best single flip in allowed
    j       ← the bit that was flipped
    tabu_until[j] ← k + L
    x       ← x'
    update x_best

return x_best
```

No temperature and no randomness. The best allowed flip is taken even when it raises the energy.

Tabu tenure board, L = 3

k	x_1	x_2	x_3	x_4
1	·	·	●	·
2	●	·	3	·
3	3	·	2	●
4	2	●	1	3
5	1	3	●	2

● bit flipped at this step 3 iterations still frozen · free



03.3

Quantum annealing

Evolution on a physical Ising machine.

Annealing schedule in Hamiltonian form

$$H(s) = -A(s)\sum_i X_i + B(s)\left(\sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j\right)$$

$s = t / t_a$ with $0 \le s \le 1$

DRIVER TERM

$$H_0 = -\sum_i X_i$$

PROBLEM TERM

$$H_P = \sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j$$

THE TWO PAULI OPERATORS

$X =$

0	1
1	0

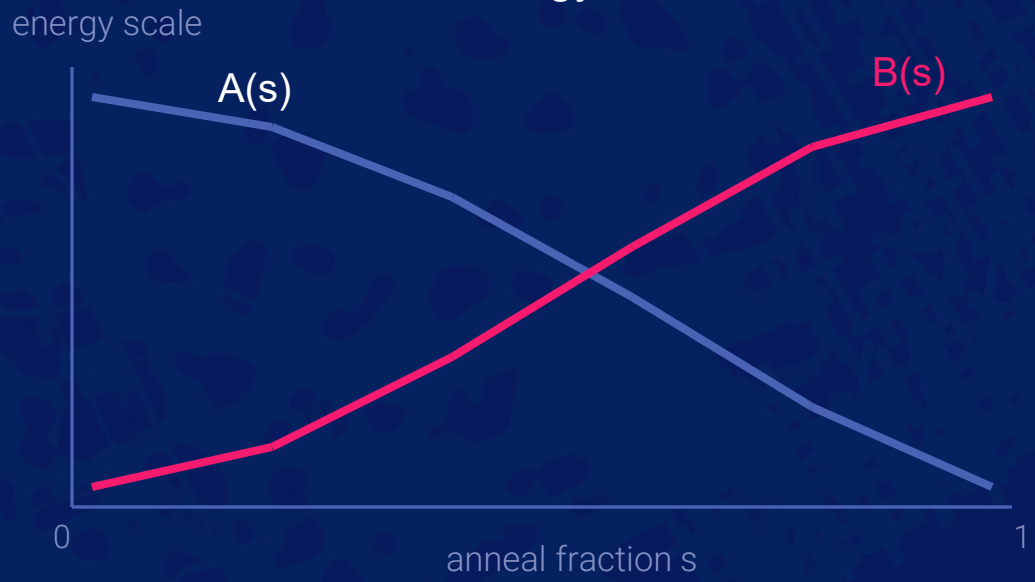
off diagonal, it flips the bit

$Z =$

1	0
0	−1

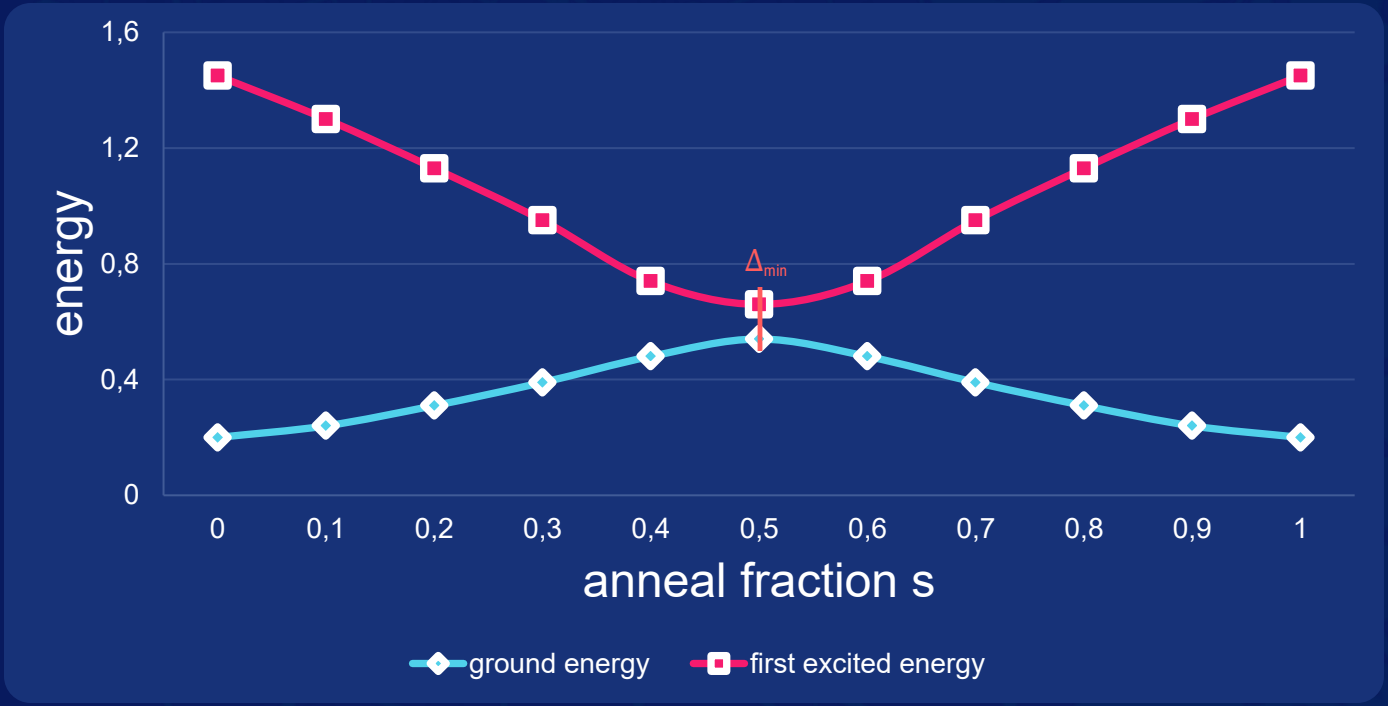
diagonal, it reads the bit as +1 or −1

Relative energy scales



Annealing changes the two weights continuously. QAOA replaces the continuous schedule with alternating cost and mixer layers.

Adiabatic theorem and minimum gap



Ground state $E_0(s)$ and first excited state $E_1(s)$

ADIABATIC STATEMENT

Start in the ground state of H_0. Sufficiently slow evolution keeps the system near that ground state.

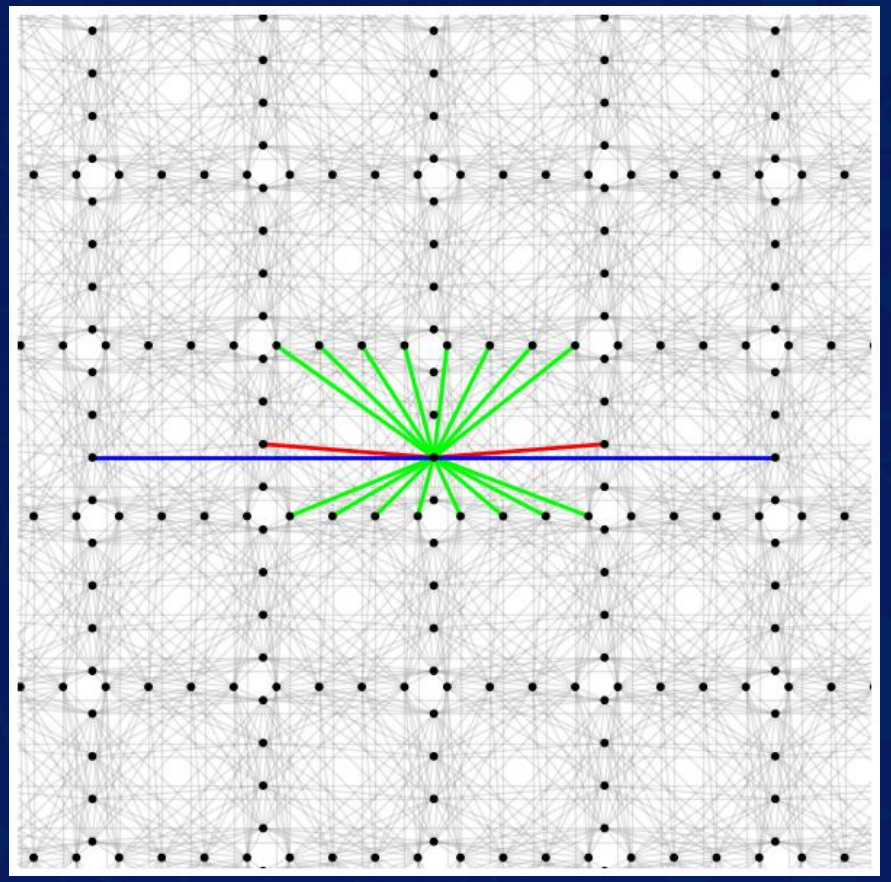
$$H(s) = A(s)H_0 + B(s)H_P$$

$$\Delta(s) = E_1(s) - E_0(s)$$
$$\Delta_{min} = \min \{\Delta(s) : 0 \leq s \leq 1\}$$

$$T \propto \Delta_{min}^{-2}$$

A smaller minimum gap requires a longer anneal.

Advantage2 hardware constraints



Zephyr couplers around one representative qubit
Image: D-Wave direct PNG. Topology: D-Wave topology documentation

4514	working qubits
40 409	working couplers
20	maximum degree
K184	ideal Z12 clique

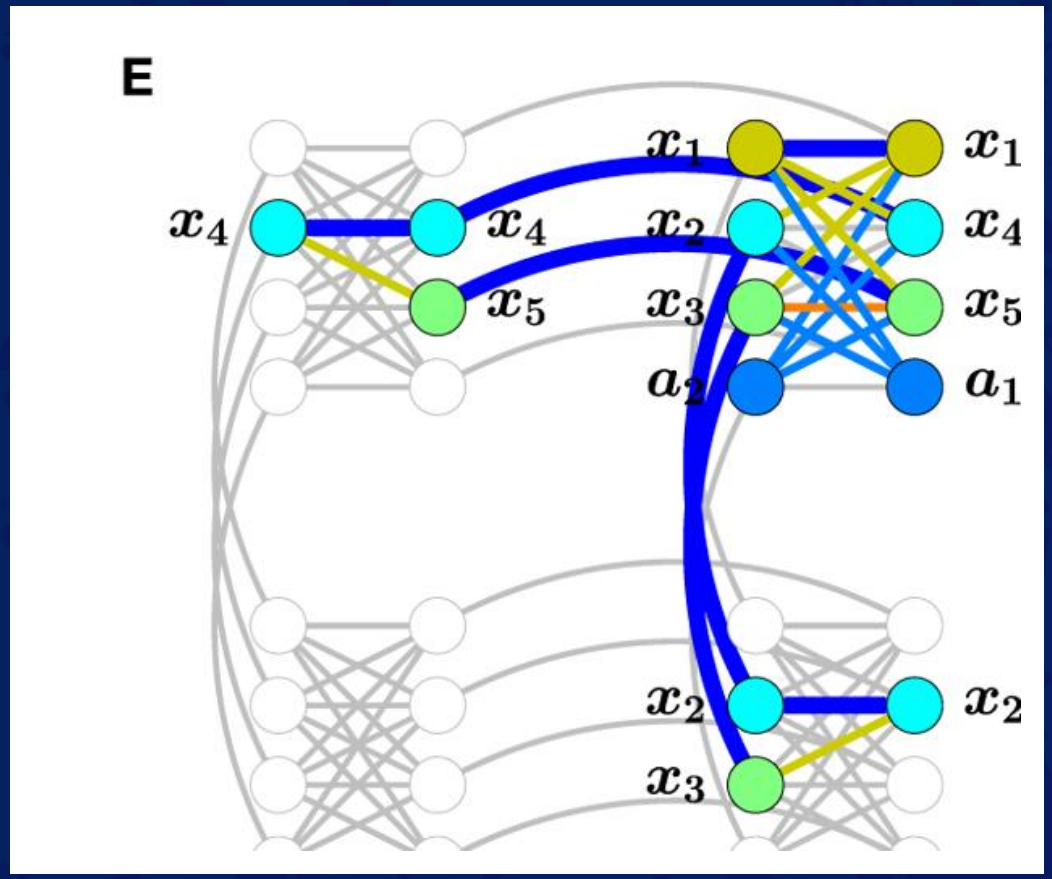
Physical qubit count overstates usable logical size

Minor embedding and chains

A logical variable may require several physical qubits

- A ferromagnetic chain represents one logical variable.
- A chain break occurs when qubits in the same chain disagree.
- The embedding search is itself a graph problem.
- Long chains consume qubits

Embedding trades physical qubits, chain strength and precision.



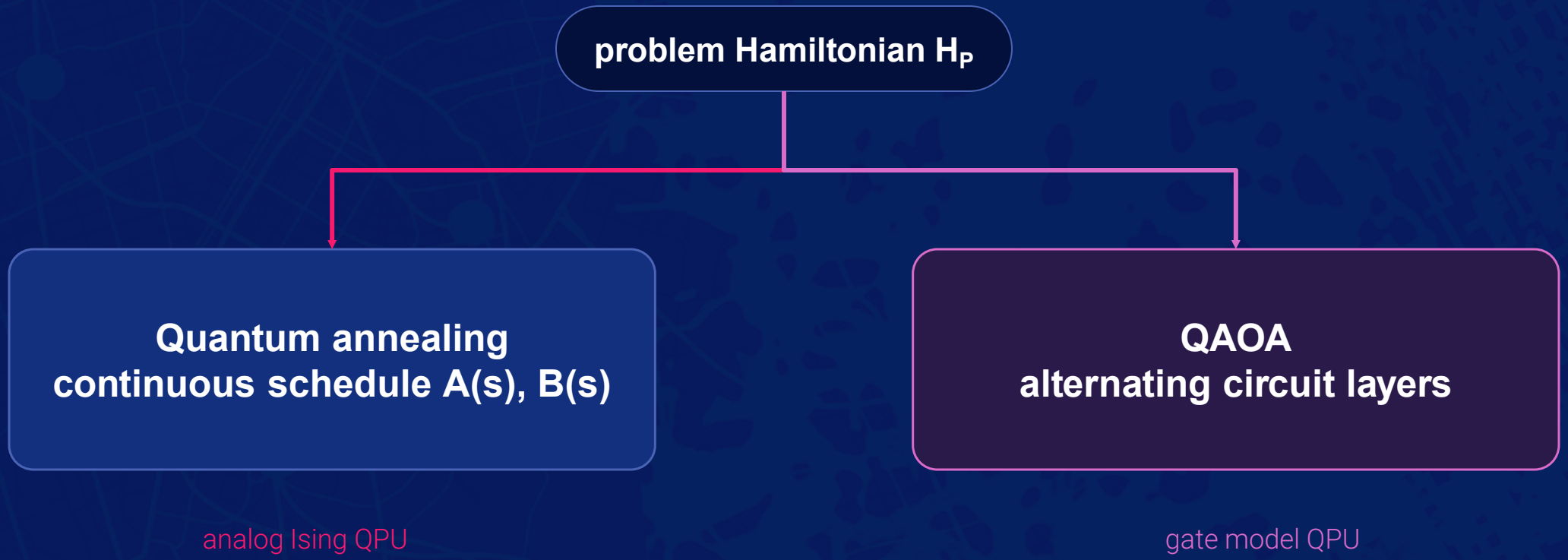
Source: Bian et al. 2016, Figure 1E

03.4

QAOA

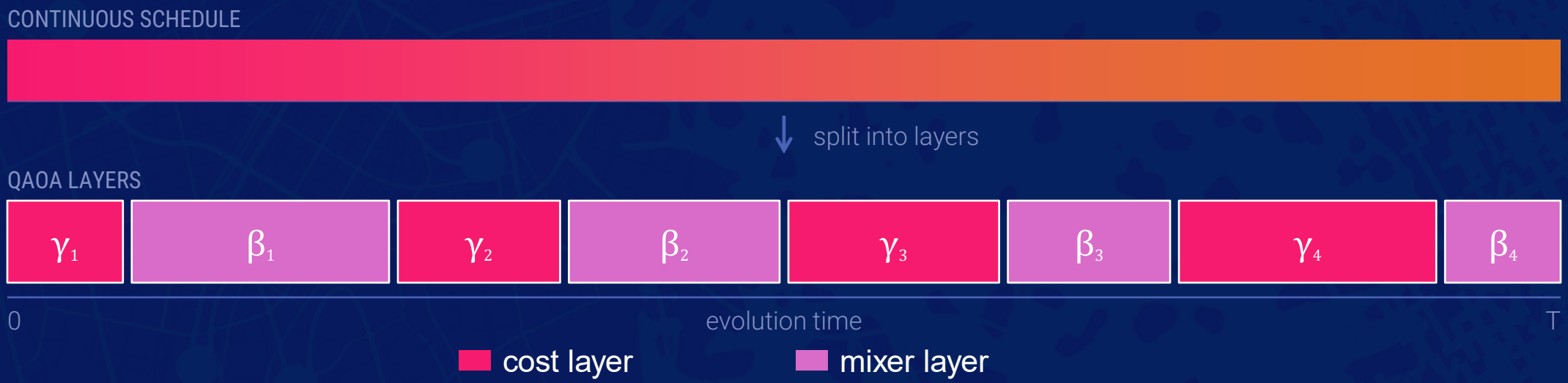
A circuit optimized by a classical outer loop.

One Hamiltonian, two machines



Same encoded objective. Different controls, errors and resource constraints.

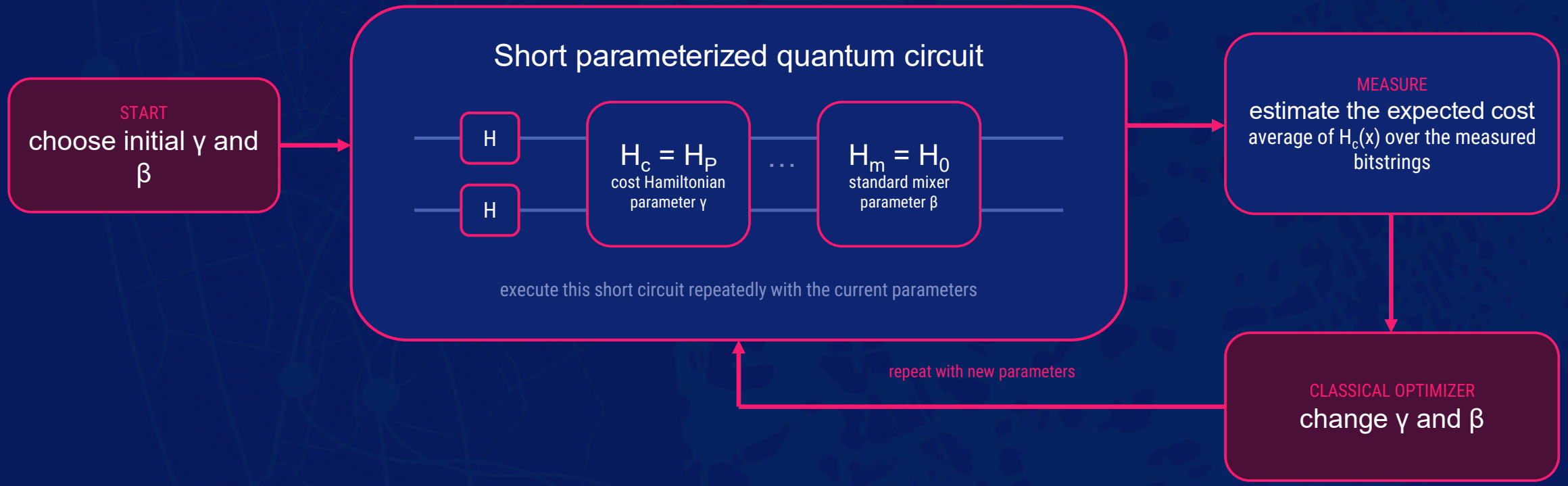
Continuous schedules and QAOA layers



Here the widths follow one fixed schedule. QAOA keeps only the alternating structure and optimizes every γ and β freely.

QAOA optimization loop

Quantum Approximate Optimization Algorithm



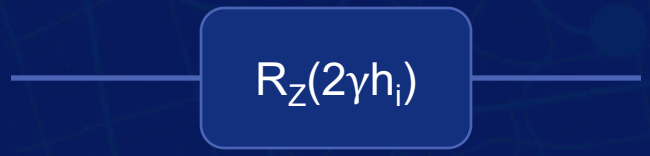
The circuit stays shallow. Repeated executions provide the samples used by the classical optimizer.

Implementing the Ising cost Hamiltonian

$$H_c = \sum_i h_i Z_i + \sum_{i<j} J_{ij} Z_i Z_j$$

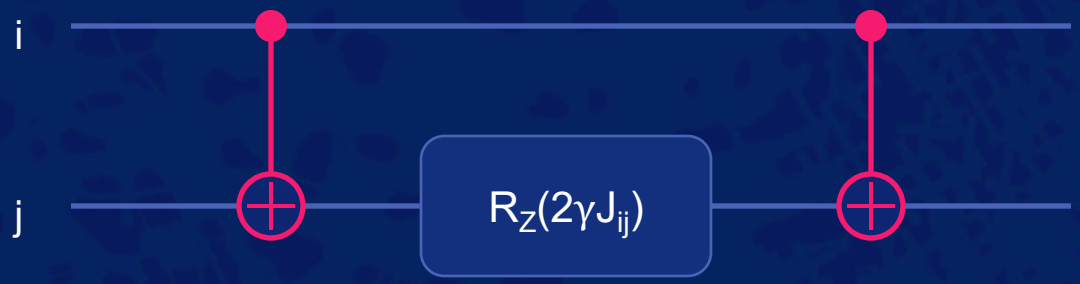
Single qubit term

$$\exp(-i \gamma h_i Z_i) = R_Z(2\gamma h_i)$$



Two qubit term

$$\exp(-i \gamma J_{ij} Z_i Z_j) = \text{CNOT}_{i \to j}\, R_Z(2\gamma J_{ij})\, \text{CNOT}_{i \to j}$$



All Z and ZZ terms commute, so their order inside the cost layer does not matter. Disjoint terms can run in parallel.

04 Benchmark and practice

Judge a solution against meaningful baselines.

Two metrics for stochastic heuristics

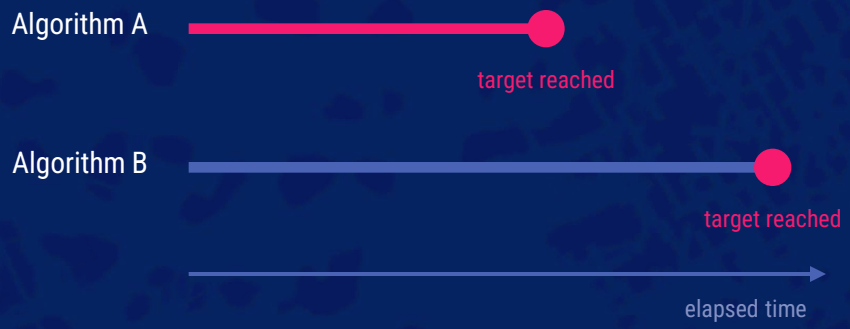
1 Quality within budget B

$$gap_r(B) = E_{best,r}(B) - E_{ref}$$

Compute one value per independent run r.
Report the median and spread.

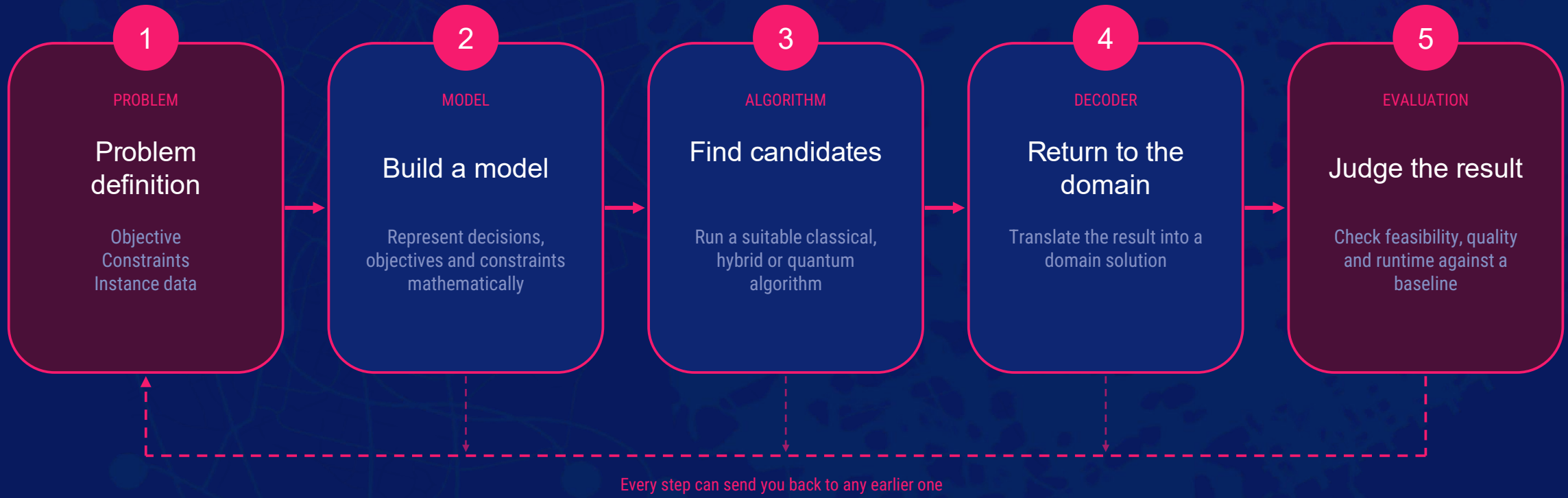
2 Time to target

Choose one target value q*.
Measure how long each algorithm takes to reach it.



These solvers are heuristics. One run is not evidence. Repeat across random seeds and problem instances.

Quick guide: from problem to evaluated solution



REFERENCES

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8 D-Wave documentation
Advantage2 solver properties
Zephyr topology
Ocean SDK documentation

9 Bian et al. (2016).
Mapping Constrained Optimization Problems to Quantum Annealing with Application to Fault Diagnosis. Frontiers in ICT 3, article 14.